Fintech Ecosystem Blueprint – Loan Wallet Settlement

Week-by-Week Implementation Checklist (Phase 1–6)

(sama seperti versi sebelumnya)

🎓Learning Companion Guide – Skill & Tools Mastery per Phase

Panduan ini membantu mengarahkan fokus pembelajaran setiap minggu agar sesuai dengan kebutuhan sistem Fintech kelas enterprise. Semua materi bisa dipelajari dengan durasi rata-rata **2 jam per hari**.

Phase 1 – Foundation (Weeks 1–4)

Goal: Bangun fondasi software architecture dan microservice environment.

Skill Focus: - Spring Boot core, configuration, dependency injection. - REST API design & validation. - Docker, Docker Compose, basic Kubernetes (pods, services, deployments). - Git branching strategy & CI/CD pipeline.

Tools & Topics: - Spring Boot 3, Spring Security 6. - PostgreSQL + Flyway. - GitHub Actions / Jenkins. - Swagger / OpenAPI. - Helm basics.

Recommended Learning Sources: - *Spring Boot in Action* – Craig Walls. - Baeldung: [spring.io/guides] - Play with Docker Labs. - YouTube channel: TechWorld with Nana (CI/CD & K8s tutorials).

Milestone Understanding:

> Kamu sudah bisa setup environment microservice lengkap dan paham CI/CD dasar.

Phase 2 – Core Wallet System (Weeks 5–8)

Goal: Kuasai transaction consistency & event-driven system.

Skill Focus: - Transactional flow, ACID & idempotency. - Kafka event streaming (producer/consumer setup). - Redis caching for idempotent request. - Error handling & distributed tracing.

Tools & Topics: - Spring Kafka, Kafka Streams. - Spring Data Redis. - Zipkin / OpenTelemetry. - AOP logging.

Recommended Learning Sources: - *Designing Data-Intensive Applications* – Martin Kleppmann. - Confluent Kafka courses (free tier). - Baeldung: Spring Kafka series.

Milestone Understanding:

> Kamu bisa membuat sistem transaksi real-time dengan jaminan konsistensi dan anti duplikasi.

Phase 3 – Loan Origination (Weeks 9–12)

Goal: Membangun proses pinjaman end-to-end & service orchestration.

Skill Focus: - DDD (Domain-Driven Design) & service boundary. - Saga pattern & eventual consistency. - Asynchronous communication & message broker. - Mock external API integration.

Tools & Topics: - Spring Cloud Stream. - Reactor (Mono/Flux). - FeignClient / WebClient. - Camunda / Temporal.io (optional advanced).

Recommended Learning Sources: - *Microservices Patterns* – Chris Richardson. - Spring Cloud documentation. - YouTube: Java Techie (Spring microservices practicals).

Milestone Understanding:

> Kamu memahami orchestrated vs. choreographed flow dan bisa membuat sistem pinjaman otomatis dengan scoring mock.

Phase 4 – Repayment & Billing (Weeks 13–16)

Goal: Menerapkan job scheduling, reconciliation, dan reliability.

Skill Focus: - Async job scheduling & retry policy. - Data reconciliation pattern. - Error recovery & DLQ (Dead Letter Queue). - Business analytics dashboard.

Tools & Topics: - Spring Batch / Quartz Scheduler. - Kafka Dead Letter Queue. - Prometheus + Grafana basic metrics. - React.js or Angular for dashboard (optional UI exposure).

Recommended Learning Sources: - Baeldung: Spring Batch guide. - Prometheus & Grafana docs. - *Building Microservices* – Sam Newman.

Milestone Understanding:

> Kamu bisa membuat proses penagihan otomatis dengan ketahanan tinggi terhadap error dan delay.

Phase 5 – Settlement & Accounting (Weeks 17–20)

Goal: Kuasai akuntansi sistem keuangan & reporting pipeline.

Skill Focus: - Double-entry bookkeeping model. - Ledger posting logic. - Data warehouse design for reporting. - Observability & metrics collection.

Tools & Topics: - Spring Data JPA advanced (native queries, projections). - Kafka Connect + Debezium (change data capture). - Grafana dashboards for financial KPIs.

Recommended Learning Sources: - *Accounting for Non-Accountants* (basics of journal/ledger). - Confluent Kafka Connect tutorials. - Grafana Labs documentation.

Milestone Understanding:

> Kamu mampu membuat sistem akuntansi otomatis dan laporan keuangan yang terhubung dengan data transaksi real-time.

Phase 6 – Risk & Compliance (Weeks 21–24)

Goal: Siapkan sistem agar aman, auditable, dan patuh regulasi.

Skill Focus: - Security best practices (API signing, encryption, rate limiting). - Audit trail & compliance design. - Threat modeling & penetration test basics. - Data masking & PII handling.

Tools & Topics: - Spring Security advanced. - Vault / KMS integration. - OWASP ESAPI / SonarQube. - Open Policy Agent (OPA) for access control.

Recommended Learning Sources: - OWASP Top 10 documentation. - *Spring Security in Action* – Laurentiu Spilca. - HashiCorp Vault tutorials.

Milestone Understanding:

> Kamu siap jadi engineer kelas enterprise dengan sistem aman dan patuh regulasi fintech (ISO/PCI DSS ready).

🔍Summary

Phase	Core Domain	Skill Keywords	Outcome
1	Foundation	Spring Boot, CI/CD, K8s	Microservice baseline ready
2	Wallet	Kafka, Redis, Consistency	Reliable wallet transaction system
3	Loan	Saga, Async, DDD	End-to-end loan flow

Phase	Core Domain	Skill Keywords	Outcome
4	Billing	Reconciliation, Retry, Metrics	Robust billing & monitoring
5	Accounting	Ledger, Reporting, Analytics	Real-time accounting pipeline
6	Compliance	Security, Audit, Vault	Regulation-ready system

Next recommended extension: **Hands-on case guide** – berisi step-by-step project per submodule (wallet, loan, billing, dsb). Mau gw tambahkan selanjutnya?